

CSE5002 Basic of CFD

Final Project. Lid-driven Cavity Flows

Due date: June 20, 2017

The lid-driven cavity problem has been used a test or validation case for incompressible Navier-Stokes equation solvers. Let's consider two-dimensional unsteady flow in a square domain with Dirichlet boundary conditions on all sides, with three stationary sides and one moving side. Based on the lid velocity U and cavity height (or width) H , Reynolds number can be defined as $Re = UH/\nu$ where ν is kinematic viscosity. The velocity of the impulsively started lid is given by a step function $u_{lid} = U$ for $t \geq 0$ and $u_{lid} = 0$ for $t < 0$.

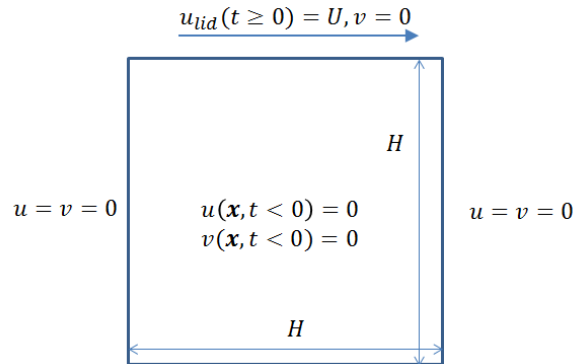


Figure 1: Schematics of flow configuration.

1. Projection method

Make your own fortran program to solve two-dimensional unsteady flow based on a projection method. Describe your projection method in detail. Note that you may use the explicit second-order Adams-Bashforth method for convection term and Crank-Nicolson scheme for viscous term. You need to use the second-order central difference scheme for spatial discretization in a uniform staggered grid.

2. Steady solutions for $Re = 100, 400$, and 1000

- (1) Compare the steady-state centerline velocities (u —velocity along the vertical centerline and v —velocity along the horizontal centerline) for $Re = 100, 400$, and 1000 with the results (see Table 1 and 2) of Ghia et al. (1982).
- (2) Plot contours of the stream function and vorticity for each Reynolds number. Compare properties of primary vortex with the results (see Table 5) of Ghia et al. (1982).
- (3) Discuss the effect of grid resolution on the steady-state solutions.
- (4) Discuss the effect of Reynolds number on the flow patterns.

3. Unsteady solutions for $Re = 100$ and 400

- (1) Plot the time history of u at the center and compare the history with the result (see Figures 7 and 8) of Dailey and Pletcher (1996).

(2) Discuss the effect of time step (or CFL number) on the unsteady solutions.

References

1. U.Ghia, K.N. Ghia, and C.T. Shin (1982) High-Re Solutions for Incompressible Flow Using the Navier-Stokes Equations and a Multigrid Method, *Journal of Computational Physics*, Vol.48, pp.387-411.
2. L.D. Dailey, and R.H. Pletcher (1996) Evaluation of Multigrid Acceleration for Preconditioned Time-Accurate Navier-Stokes Algorithms, *Computers & Fluids*, Vol.25, pp.791-811.